



PROJECT-BASED LEARNING

PHASE-I EVALUATION

College Event Registration Platform

A Simple Event Registration and Management System

Team ID: DSCPP-III-2026-T188

Department of Computer Science & Engineering

Graphic Era (Deemed to be University), Dehradun

Academic Session 2026–27

Team & Mentor Details



Team Information

Team ID: DSCPP-III-2026-T188

Semester: 3rd Semester

Section: AIML2 , AIML3 , AIML4

Domain: Data Structures and C++

Team Members

1. Nikunj Singhal (Team Lead) — 2510370169
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3. Harshit Kashyap — 2510011536

Mentor

Dr. Prabhdeep Singh

Interactions completed: _____

Problem Statement

College events like workshops, competitions, seminars and cultural activities are usually managed through different forms, spreadsheets or separate platforms because of this students may find it difficult to know about upcoming events, check available seats and keep track of their registrations. Organizers also have to manage participant details, event capacity and registration records manually.

Motivation

- Event information is scattered across different platforms
- Difficult to track seats and registration status
- Manual management takes time for organizers

Target Users / Stakeholders

- College students registering for events.
- Event organizers and coordinators.
- College administration managing events.

Project Objectives

1. Develop a simple college event registration system using C++.
2. Manage events, students and registrations efficiently.
3. Allow easy event search, registration and cancellation.
4. Apply OOP and Data Structures concepts in the project.
5. Provide a simple console-based interface for users.

Key Objective: To help students easily register for college events and help organizers manage events and registrations.

Proposed Approach

1. **Identify the main entities** — Student, Event, Organizer and Registration.
2. Create classes and functions for each part of system.
3. Use linked lists ,queues ,stacks to manage records.
4. Add registration ,cancellation ,searching and sorting functions.
5. Test each module and combine them into final C++ system.

Key Features

- Event creation and management.
- Student registration and cancellation.
- Seat capacity management.
- Event search and filtering.
- View registration records and basic reports.

Clearly provides a single platform for students to register for events and for organizers to manage events and registrations.

Integration of PBL Course Concepts



Course	Concepts Applied	Project Module
Data Structures in C++	Linked Lists, Queues, Stacks, Searching & Sorting	Event and Registration Management
OOPs with C++	Classes, Objects, Encapsulation, Inheritance & Polymorphism	Student, Event, Organizer and Registration

Requirement: OOPs and Data Structures concepts are combined to manage events, student registrations, seats and records.



Major Components

- Student Management
- Event Management
- Registration Management

Data / Control Workflow

- Student → Search Event → Register / Cancel
- Organizer → Create / Update / Delete Event
- System → Check Seats → Update Registration Status

Technology Stack

Programming: C++

Data Structure: Linked List, Queue, Stack

IDE: Visual Studio Code

Version Control: Git / GitHub

Development Methodology

1. Requirement Analysis
2. System Design (object model & architecture)
3. Implementation (milestone-wise, Week 1–8)
4. Testing (event registration, cancellation & capacity)
5. Evaluation & Report

C++ supports Data Structures and OOP concepts, while the C++ Standard Library provides useful tools for managing data. VS Code, GitHub, and the step-by-step methodology make development and testing easier.



Work Done So Far

- College event registration requirements identified.
- Student, organizer and event requirements prepared.
- C++ system architecture and basic workflow prepared.
- Main classes and modules planned.
- Basic data structures and registration flow prepared.

Team Contributions

Member	Role	Contribution
Nikunj Singhal	Team Lead	Planning, coordination, system development and DSA
Bhargav Istwal	Developer/Docs	C++ development, OOP, DSA and documentation
Harshit Kashyap	Testing	Testing and system verification

Our Roadmap

1. Phase-I: Requirements & System Design
2. Phase-II: C++ Development & DSA Implementation
3. Phase-III: Testing, Final Integration & Presentation

Expected Results

- Working C++ college event registration system
- Easy event search and registration
- Proper seat and registration management
- Organized student and event records

References

1. cppreference – C++ Classes and Objects
2. cppreference — Inheritance and Polymorphism
3. cppreference – C++ Data Structures and Containers
4. GeeksforGeeks – Data Structures and Algorithms

Thank You

Questions & Suggestions